

ASSESSING THE IMPACT OF TEMPORAL RESOLUTION ON ANIMAL BEHAVIOURAL INFERENCE



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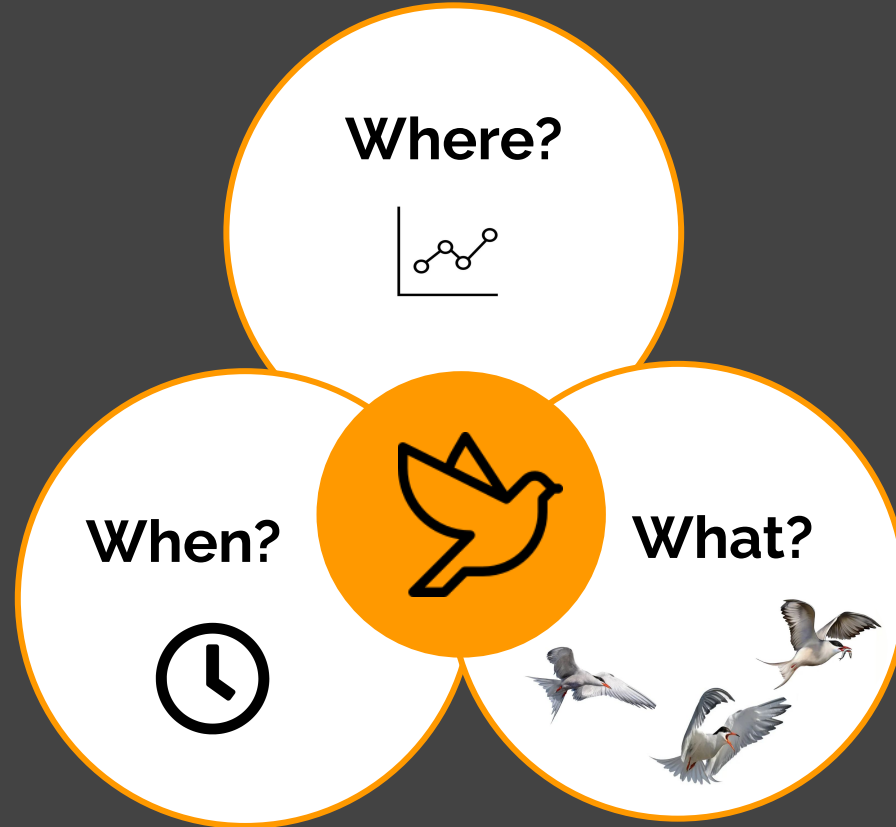
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DR GAIL ROBERTSON

– Background



— Question



How do we observe movements at discrete-time with little or no loss in information to determine behaviour classification



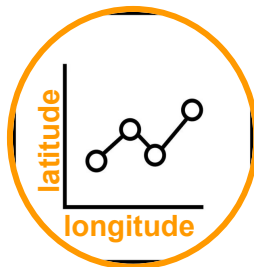
– Goal

Assess effect of sampling frequency on animal behavioural inference **using:**

- Thinning approach
- Nyquist-Shannon Sampling Theorem

— Nyquist-Shannon sampling theorem (NSST)

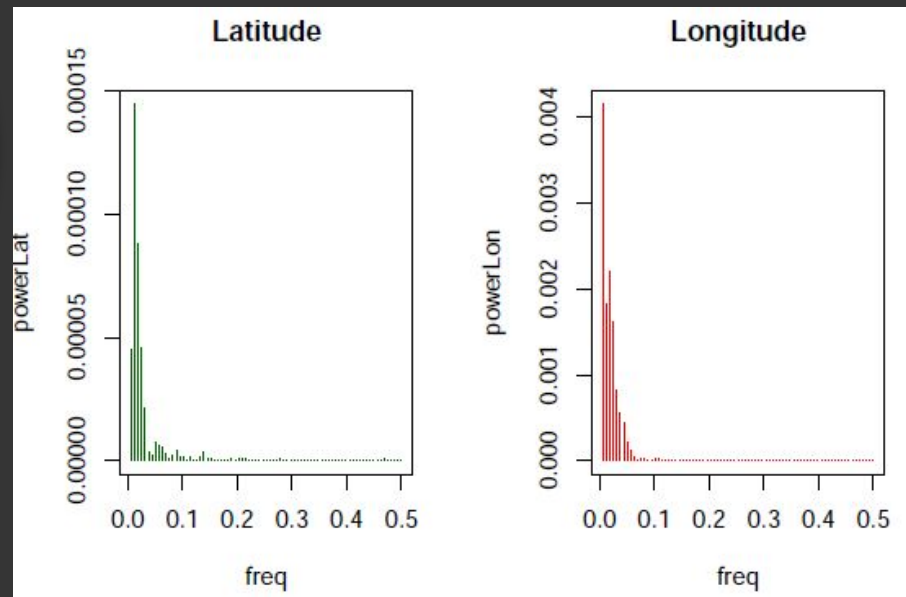
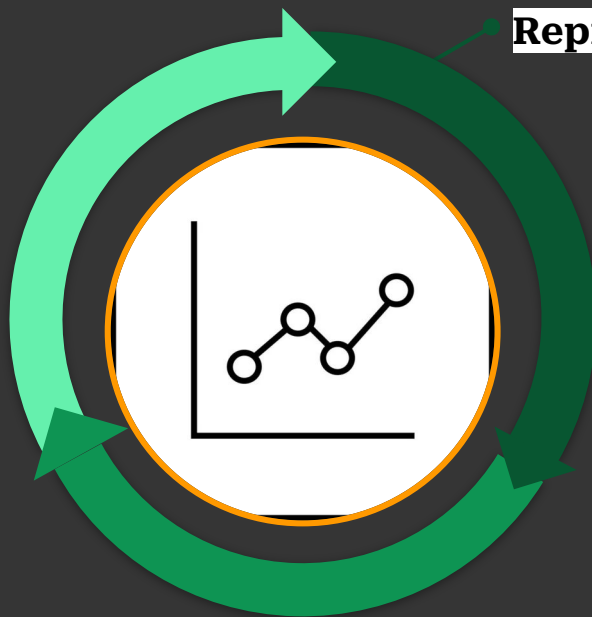
- Describes how to sample a continuous-time signal (or subsample a discrete-time signal) in a way that there is little or no information loss
- Useful for selecting frequencies to subsample movement trajectories



— NSST workflow

Step 1.

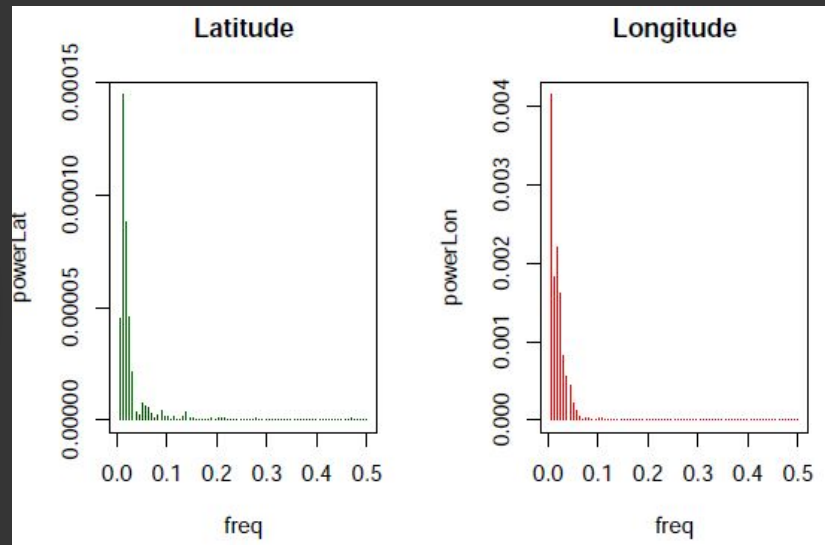
Represent movement tracks in frequency domain



— NSST workflow

Step 1.

Represent movement tracks in frequency domain

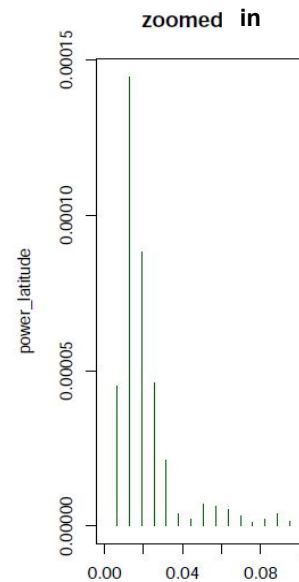
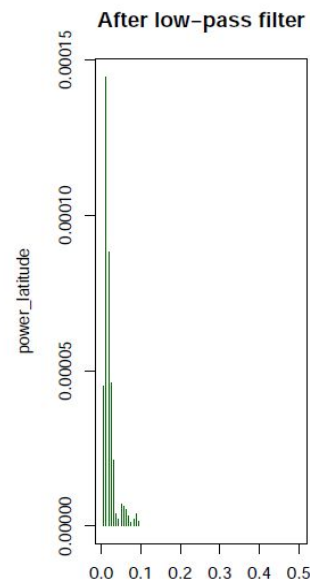
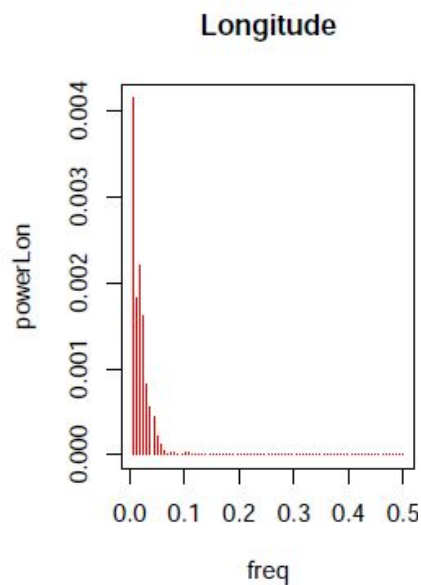
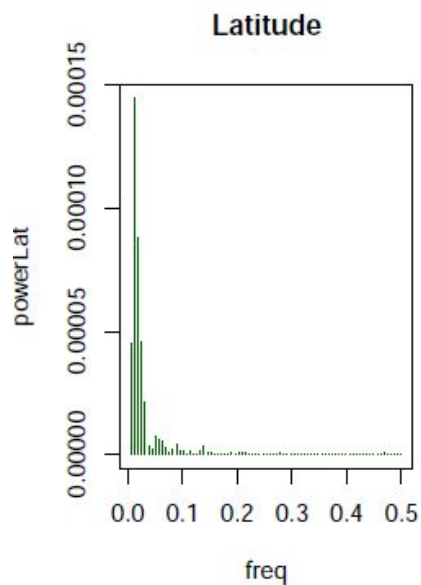


Step 2.

Select a cut-off frequency, **B'**, for subsampling

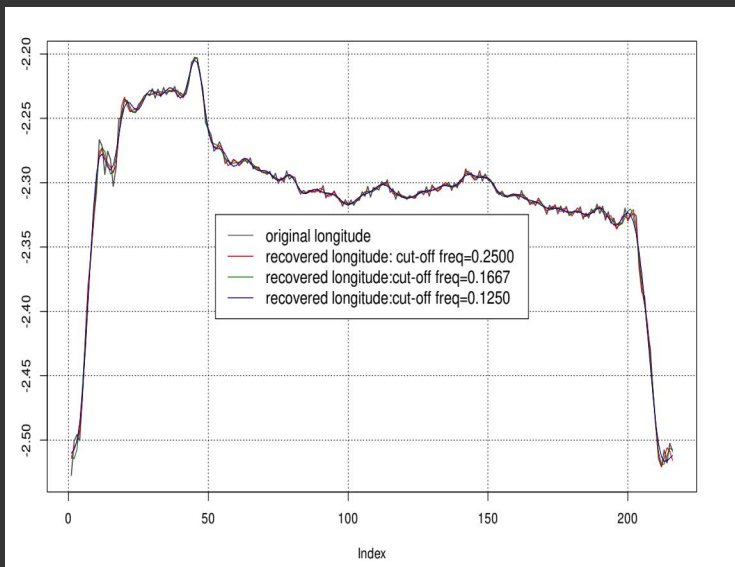
— NSST workflow ~ low pass filter

- Discard high frequencies at a cut-off frequency of $B' < 0.5$.
- Then subsample, working at a $fs' = B' < 0.5$



— NSST workflow

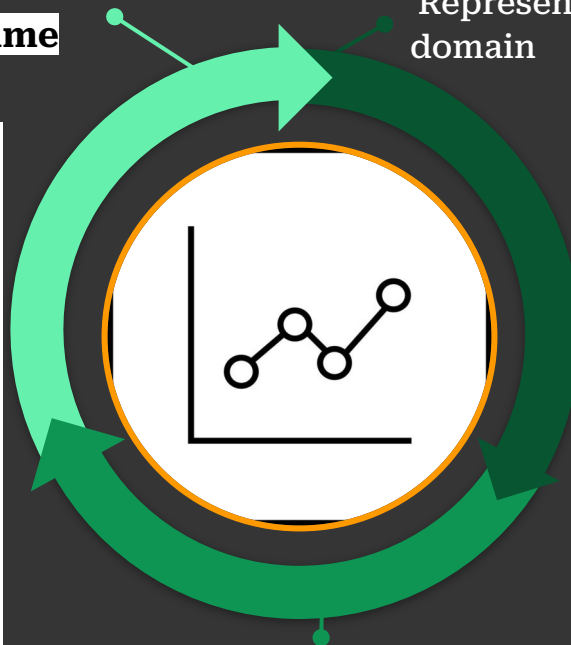
Recover filtered movement tracks in time domain



Step 3.

Step 1.

Represent movement tracks in frequency domain



Step 2.

Select a cut-off frequency, B' , for subsampling

– Goal

- **Thinning approach**

Investigate how behavioural inference is impacted due to:

1. having less information in the trajectory
2. Each segment not being exclusively one state

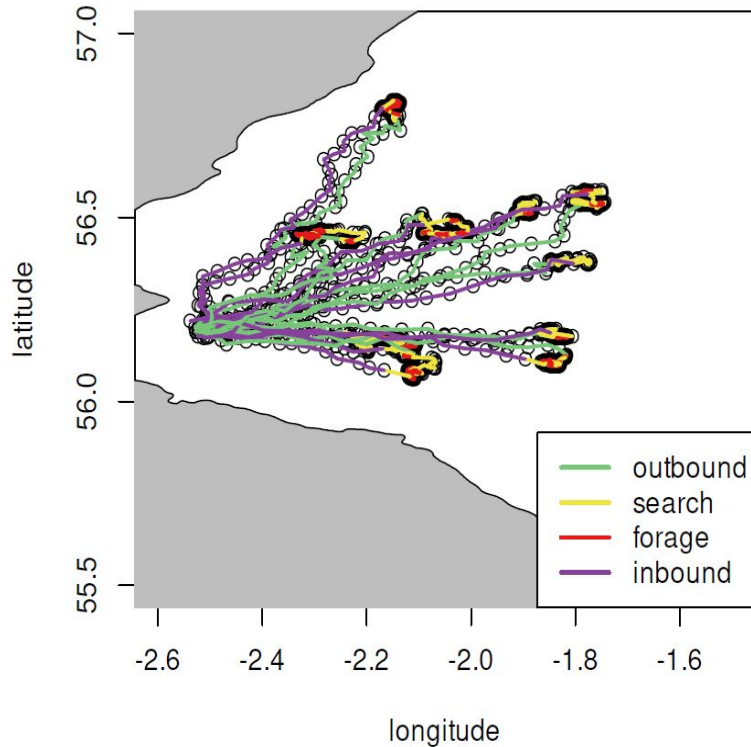
e.g



- **Nyquist-Shannon Sampling Theorem**

Assess how the chosen cut-off frequency, **B'**, impacts behavioural inference

— Example : simulation study



- **Thinning:** 10, 15, and 20 mins

Metric: accuracy and f1-score

- **NSST:** Quantiles to select cut-off frequency

Metric: accuracy and f1-score

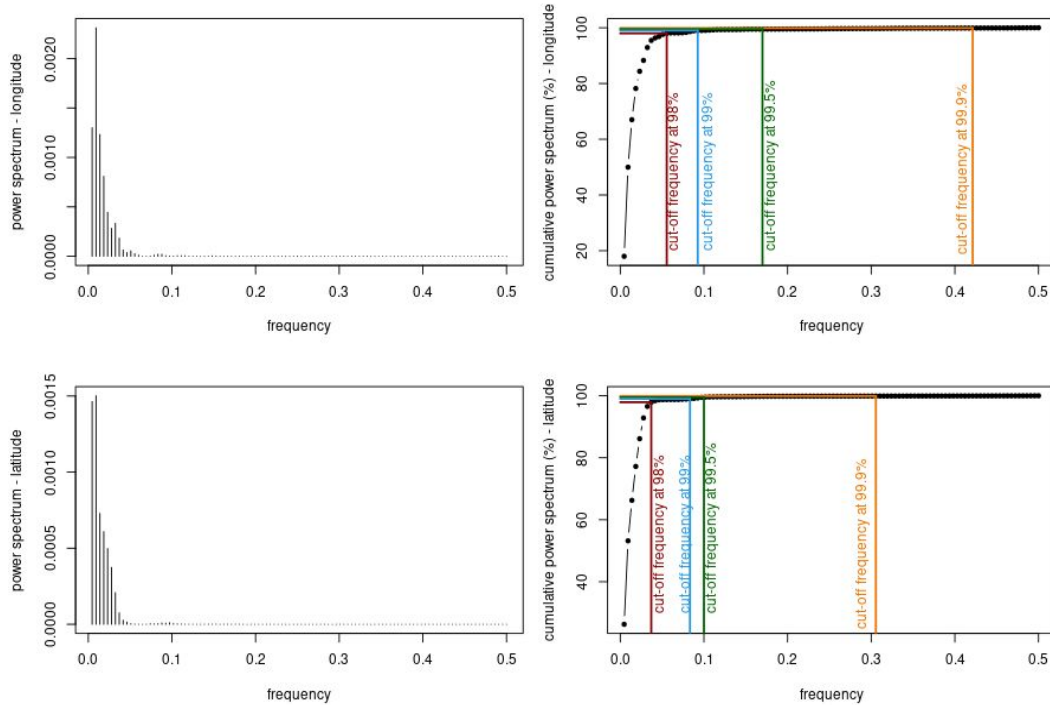
– Result ~ Thinning approach

Metric 2: count as correct the decoded state if it was originally present in the **segment**.

S	S	F	F	F	S	S	S	S	S	S	S
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	accuracy(%)	travel	F1-score(%)	
			forage	search
Original track				
Metric 1	95.83	96.88	97.78	83.02
10 mins thin				
Metric 1	90.74	96.77	94.55	50.00
Metric 2	92.59	96.77	95.76	60.00
15 mins thin				
Metric 1	90.28	90.00	95.50	46.15
Metric 2	94.44	94.74	97.25	75.00
20 mins thin				
Metric 1	87.04	87.50	92.86	25.00
Metric 2	94.44	100.00	96.47	57.14

– Result ~ NSST Approach



	accuracy (%)	travel	forage	search
Original track	95.83	98.17	96.67	92.43
Q-99.9%	94.44	98.17	92.35	81.22
Q-99.5%	88.89	93.48	87.27	71.12
Q-99%	80.56	93.48	83.86	73.36
Q-98%	53.70	87.50	71.96	42.91

– Take home

- **Thinning approach**

Less accuracy when unable to account for more than one exclusive behaviour in each segment

- **NSST approach**

Cut-off frequency chosen based on high quantile of the spectrum distribution is preferable

— References

Akeresola et al. (2024). [Validating hidden Markov models for seabird behavioural inference](#), *Ecology and Evolution*. doi/10.1002/ece3.11116

Shannon, C. E. (1949) [Communication in the presence of noise](#), *Proceedings of the I.R.E.* 37(10-21). doi.org/10.1109/JRPROC.1949.232969

Nathan et al. (2022). [Big-data approaches lead to an increased understanding of the ecology of animal movement](#). *Science*. 375(6582). doi:10.1126/science.abg1780

Many thanks to

Supervisory team



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Thanks for your attention

